PSEB Class 10 Science

Chapter 1: Chemical Reactions and Equations

Objective Questions with Answers | English Medium

Practice set: 30 MCQs, 15 Fill in the Blanks, and 20 True/False questions. These are practice questions, not an official guaranteed exam list.

Part A: Multiple Choice Questions (MCQs) — 30 Questions

Q1. Which of the following is a chemical change?

A. Melting of ice
B. Boiling of water
C. Rusting of iron
D. Breaking of glass

Answer: C. Rusting of iron

Q2. The substances that undergo a chemical reaction are called:

A. Products
B. Reactants
C. Catalysts
D. Elements

Answer: B. Reactants

Q3. The substances formed during a chemical reaction are called:

A. Reactants
B. Products
C. Mixtures
D. Solvents

Answer: B. Products

Q4. Which gas is released when zinc reacts with dilute hydrochloric acid?

A. Oxygen
B. Nitrogen
C. Hydrogen
D. Carbon dioxide

Answer: C. Hydrogen

Q5. A balanced chemical equation has:

A. Equal numbers of molecules on both sides
B. Equal numbers of atoms of each element on both sides
C. Equal masses of every substance
D. Equal numbers of compounds on both sides

Answer: B. Equal numbers of atoms of each element on both sides

Q6. Which gases are released when lead nitrate is heated?

A. Oxygen only
B. Nitrogen dioxide and oxygen
C. Hydrogen
D. Carbon monoxide

Answer: B. Nitrogen dioxide and oxygen

Q7. The reaction 2Mg + O2 -> 2MgO is an example of:

A. Decomposition reaction
B. Combination reaction
C. Displacement reaction
D. Double displacement reaction

Answer: B. Combination reaction

Q8. Which of the following is a decomposition reaction?

A. $CaO + H_2O \rightarrow Ca(OH)_2$
B. $2AgCl \rightarrow 2Ag + Cl_2$
C. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
D. $HCl + NaOH \rightarrow NaCl + H_2O$
Answer: B. $2AgCl \rightarrow 2Ag + Cl_2$

Q9. The decomposition of silver chloride takes place in the presence of:

A. Sunlight
B. Water
C. Nitrogen
D. Hydrogen
Answer: A. Sunlight

Q10. Which of the following is a displacement reaction?

A. $CaO + H_2O \rightarrow Ca(OH)_2$
B. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
C. $2H_2 + O_2 \rightarrow 2H_2O$
D. $CaCO_3 \rightarrow CaO + CO_2$
Answer: B. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

Q11. When an iron nail is dipped in copper sulphate solution, the solution changes from blue to:

A. Red
B. Green
C. Yellow
D. Colourless
Answer: B. Green

Q12. Which of the following is a double displacement reaction?

A. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
B. $CaCO_3 \rightarrow CaO + CO_2$
C. $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$
D. $2Mg + O_2 \rightarrow 2MgO$
Answer: C. $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$

Q13. The white precipitate formed when barium chloride reacts with sodium sulphate is:

A. Sodium chloride
B. Barium sulphate
C. Sodium sulphate
D. Barium chloride
Answer: B. Barium sulphate

Q14. Oxidation involves:

A. Gain of oxygen
B. Loss of oxygen
C. Gain of nitrogen
D. Loss of hydrogen only
Answer: A. Gain of oxygen

Q15. Reduction involves:

A. Gain of oxygen
B. Loss of oxygen
C. Gain of carbon dioxide
D. Gain of nitrogen
Answer: B. Loss of oxygen

Q16. A reaction in which heat is released is called:

A. Endothermic reaction
B. Exothermic reaction
C. Decomposition reaction
D. Precipitation reaction

Answer: B. Exothermic reaction

Q17. Which of the following is an exothermic reaction?

A. Photosynthesis
B. Thermal decomposition of calcium carbonate
C. Burning of natural gas
D. Thermal decomposition of lead nitrate

Answer: C. Burning of natural gas

Q18. Photosynthesis is an example of:

A. Exothermic reaction
B. Endothermic reaction
C. Displacement reaction
D. Precipitation reaction

Answer: B. Endothermic reaction

Q19. The black coating formed on copper when heated in air is:

A. Copper sulphate
B. Copper oxide
C. Copper carbonate
D. Copper chloride

Answer: B. Copper oxide

Q20. The chemical formula of rust is commonly represented as:

A. FeO
B. FeS
C. $Fe_2O_3.xH_2O$
D. $FeCO_3$

Answer: C. $Fe_2O_3.xH_2O$

Q21. Rancidity is caused by the oxidation of:

A. Proteins
B. Fats and oils
C. Water
D. Minerals

Answer: B. Fats and oils

Q22. Which gas is necessary for combustion?

A. Nitrogen
B. Oxygen
C. Hydrogen
D. Helium

Answer: B. Oxygen

Q23. Which of the following helps prevent rancidity?

A. Keeping food open
B. Adding antioxidants
C. Exposing food to sunlight
D. Keeping food in a warm place

Answer: B. Adding antioxidants

Q24. Which of the following is a combination reaction?

A. CaO + H2O -> Ca(OH)2
B. 2AgCl -> 2Ag + Cl2
C. Fe + CuSO4 -> FeSO4 + Cu
D. CaCO3 -> CaO + CO2
Answer: A. CaO + H2O -> Ca(OH)2

Q25. When ferrous sulphate crystals are heated strongly, the solid residue contains:

A. Iron oxide
B. Copper oxide
C. Zinc oxide
D. Calcium oxide
Answer: A. Iron oxide

Q26. Which symbol is used to represent a gas in a chemical equation?

A. (s)
B. (l)
C. (g)
D. (aq)
Answer: C. (g)

Q27. The symbol (aq) in a chemical equation means:

A. Solid
B. Liquid
C. Aqueous solution
D. Gas
Answer: C. Aqueous solution

Q28. Which of the following represents a balanced chemical equation?

A. H2 + O2 -> H2O
B. 2H2 + O2 -> 2H2O
C. H2 + 2O2 -> H2O
D. 2H2 + 2O2 -> H2O
Answer: B. 2H2 + O2 -> 2H2O

Q29. The reaction CuO + H2 -> Cu + H2O involves:

A. Oxidation of copper oxide
B. Reduction of copper oxide and oxidation of hydrogen
C. Reduction of hydrogen
D. No redox change
Answer: B. Reduction of copper oxide and oxidation of hydrogen

Q30. Which of the following is a precipitation reaction?

A. 2H2 + O2 -> 2H2O
B. Zn + 2HCl -> ZnCl2 + H2
C. Pb(NO3)2 + 2KI -> PbI2 + 2KNO3
D. CaCO3 -> CaO + CO2
Answer: C. Pb(NO3)2 + 2KI -> PbI2 + 2KNO3

Part B: Fill in the Blanks — 15 Questions

1. A chemical reaction can be represented by a chemical _____.
2. The substances present before a chemical reaction are called _____.
3. The substances formed in a chemical reaction are called _____.
4. A balanced chemical equation obeys the law of conservation of _____.
5. Magnesium burns in oxygen to form magnesium _____.
6. The reaction in which two or more substances combine to form a single product is called a _____ reaction.
7. The breakdown of a single compound into simpler substances is called a _____ reaction.
8. The addition of oxygen to a substance is called _____.
9. The removal of oxygen from a substance is called _____.
10. A reaction that releases heat is called an _____ reaction.
11. A reaction that absorbs heat is called an _____ reaction.
12. The black coating formed on heated copper is copper _____.
13. The blue colour of copper sulphate solution changes to _____ when an iron nail is placed in it.
14. The white precipitate formed by barium chloride and sodium sulphate is _____.
15. The oxidation of fats and oils that causes an unpleasant smell and taste is called _____.

Answers: 1. Equation 2. Reactants 3. Products 4. Mass 5. Oxide 6. Combination 7. Decomposition 8. Oxidation 9. Reduction 10. Exothermic 11. Endothermic 12. Oxide 13. Green 14. Barium sulphate 15. Rancidity

Part C: True or False — 20 Questions

1. Burning of magnesium is a chemical change. Answer: True

2. A chemical equation must always have equal numbers of molecules on both sides. Answer: False

3. The total number of atoms of each element remains the same in a balanced chemical equation. Answer: True

4. Rusting of iron is a physical change. Answer: False

5. A combination reaction produces a single product from two or more reactants. Answer: True

6. Decomposition reactions require only heat and never electricity or light. Answer: False

7. Silver chloride turns grey in sunlight. Answer: True

8. An iron nail dipped in copper sulphate solution causes the blue solution to turn green. Answer: True

9. In a displacement reaction, a more reactive element can displace a less reactive element from its compound. Answer: True

10. Oxidation means the gain of oxygen. Answer: True

11. Reduction means the gain of oxygen. Answer: False

12. Burning of fuels is generally an exothermic reaction. Answer: True

13. Photosynthesis absorbs energy from sunlight. Answer: True

14. Rancidity can be reduced by storing food in airtight containers. Answer: True

15. Barium sulphate is a white precipitate. Answer: True

16. In a double displacement reaction, ions are exchanged between two compounds. Answer: True

17. Hydrogen gas is released when zinc reacts with dilute hydrochloric acid. Answer: True

18. The formula of magnesium oxide is MgO. Answer: True

19. Rusting of iron occurs without oxygen or moisture. Answer: False

20. Antioxidants can help prevent rancidity. Answer: True

Quick Revision: Important Chemical Equations

Reaction	Balanced equation
Burning of magnesium	2Mg + O2 -> 2MgO
Formation of water	2H2 + O2 -> 2H2O
Zinc with hydrochloric acid	Zn + 2HCl -> ZnCl2 + H2
Thermal decomposition of calcium carbonate	CaCO3 --heat--> CaO + CO2
Iron with copper sulphate	Fe + CuSO4 -> FeSO4 + Cu
Silver chloride in sunlight	2AgCl --sunlight--> 2Ag + Cl2
Barium chloride with sodium sulphate	BaCl2 + Na2SO4 -> BaSO4 + 2NaCl
Copper oxide with hydrogen	CuO + H2 -> Cu + H2O

Total in this practice set: 30 MCQs + 15 Fill in the Blanks + 20 True/False = 65 questions.

Note: These are revision questions. Check your current PSEB textbook and teacher's instructions for the latest syllabus.